Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · Distributed Systems · AWS · Large-Scale Data Pipelines · Architecture · ML/LLM · 9yr Production

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PROFESSIONAL SUMMARY

Senior software engineer with 9+ years of production experience designing, building, and owning large-scale distributed systems, data pipelines, and enterprise inventory workflows - the reliability, scalability, and architectural ownership standards AWS's Supply Chain Management team requires. At Entrata, I owned the HUD federal compliance pipeline for 7 years: a Python and AWS system processing inventory and warehouse data (federal source records → 180,000 domain records) that drives financial calculations and eligibility decisions for 100K+ housing units - where an incorrect record means incorrect payments and regulatory violations. This is the same high-stakes inventory accuracy problem AWS SCM solves for physical infrastructure at global scale. I led the architecture and redesign of multiple production systems: Docker microservices redesign (50% scalability improvement, 30% deployment time reduction), AWS S3 document migration (100% PHI compliance, zero data loss), RabbitMQ event-driven async processing system (retry logic, idempotency, dead-letter handling), and 200+ SQL optimizations (30% performance gain) - all production systems I owned from design through ongoing operation. I also build and deploy ML and LLM systems in production: AgentProbe (evaluation platform with transformer-based LLM integration) and LeaseI (multi-agent reasoning system with RAG) - addressing the ML/LLM fundamentals preferred qualification. I mentor engineers, lead code reviews, and deliver cross-functional projects from design to production at 99.5% uptime. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Systems: Python (9yr), SQL, PL/SQL, PHP, TypeScript, JavaScript, Bash, FastAPI, REST APIs, Linux (production 9yr), Git, Distributed Systems, System Design, Architecture Patterns, Reliability Engineering

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Microservices, RabbitMQ (event-driven async, retry logic, idempotency), Redis, PostgreSQL, MySQL, Oracle, GitHub Actions CI/CD, Automated Testing, Performance Engineering, Scalability, Observability, ETL Pipelines, Data Modeling

ML/LLM & Data: LangGraph, LangChain, Multi-Agent Systems, LLM Evaluation, RAG Pipelines, ChromaDB, Transformer Architecture, PyTorch, scikit-learn, NLP, BERT, spaCy, Statistical Modeling, Data Pipelines, Pandas, NumPy, Agile/Scrum, Technical Mentoring

PROJECTS

AgentProbe: Large-Scale Distributed Evaluation Platform - ML/LLM

Distributed Systems · ML/LLM · Scalability · Open Source

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Scalable distributed evaluation platform for ML/LLM systems - built with the same reliability, observability, and scalability patterns production supply chain infrastructure requires. **Architecture:** multi-provider LLM routing with clean API abstractions; ChromaDB vector database with chunking optimization; 3-agent LangGraph orchestration pipeline. **ML/LLM integration:** transformer-based LLM APIs (Anthropic, OpenAI), embedding models, evaluation harness with LLM-as-Judge scoring. **Reliability:** CI-integrated regression gate blocking quality degradation; 5-metric observability engine; zero regressions across 1,240+ test cases. Enterprise-deployable, open-source.

LeaseI: Multi-Agent Reasoning System - Inventory & Data Orchestration

Multi-Agent · Distributed · Data Pipelines · Production

Python · LangGraph (8-agent) · FastAPI · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Distributed multi-agent system orchestrating data retrieval, analysis, and decision workflows across multiple data sources - demonstrating the distributed system design and data orchestration patterns AWS SCM applies at inventory scale. **System design:** 8-agent LangGraph StateGraph with conditional routing and dynamic task delegation; shared long-term data store (SQLite knowledge repository) across all agents and sessions; fault-tolerant escalation paths. **Data pipelines:** OCR ingestion, section-aware extraction, multi-source retrieval with cosine reranking. 93.6% precision across 1,240 test cases; CI deployment gate throughout.

AI on FHIR: Cloud-Native Distributed System - AWS & Scalable APIs

Cloud-Native · AWS · Scalable APIs · Architecture

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Cloud-native distributed system integrating ML inference with enterprise REST APIs - containerized with Docker for scalable deployment; HIPAA-compliant RBAC with authentication and data isolation; NLP-powered query translation reducing manual retrieval time ~60%. Demonstrates full software development lifecycle ownership: design, implementation, testing, deployment, and compliance validation across a distributed cloud architecture.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production Systems Architecture & Delivery

- Architected and delivered a production distributed AI system (Python, LangGraph, FastAPI) end-to-end - requirements, system design, implementation, testing, and production deployment as sole technical owner; designed fault-tolerant workflows with retry logic, validation gates, and human escalation paths for high-stakes decisions.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977) on WHO/UK datasets (190+ countries); designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and distributed systems design.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Large-scale distributed systems architecture and ownership** - owned the HUD federal compliance pipeline for 7 years (Python, AWS Glue/S3/RDS; federal inventory data → 180,000 domain records driving financial calculations for 100K+ housing units); led Docker microservices architecture redesign (50% scalability improvement, 30% deployment time reduction); designed RabbitMQ event-driven async system with retry logic, idempotency guarantees, and dead-letter handling; 99.5% uptime across multi-tenant SaaS.
- **AWS infrastructure and data pipeline engineering** - led AWS S3 document migration for Affordable Housing module (100% PHI compliance, zero data loss, full metadata validation); AWS S3/EC2/RDS/Glue production data pipelines; 200+ SQL optimizations (30% performance gain) through execution plan analysis, indexing, and partitioning; Python/Bash automation pipelines eliminating 80% of manual operational workload; Redis caching.
- **Technical leadership and full SDLC ownership** - 20+ full-stack production systems shipped across 120+ Agile sprints; GitHub Actions CI/CD with automated testing throughout; mentored 5+ engineers on system design, scalability, and production practices; 300+ peer code reviews; internal debugging tools reducing triage time 40%; cross-functional delivery with product, compliance, and operations stakeholders.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; inventory data pipeline for 1.6M records across 5 program years; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017